

My Project

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Chapter 1

Hierarchical Index

1.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

CustomVector< T >	7
Human	14
Student	17

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CustomVector< T >	7
Human	14
Student	17

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

include/create-student.h	23
include/custom-vector-push-back-test.h	25
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Chapter 4

Class Documentation

4.1 CustomVector< T > Class Template Reference

```
#include <custom-vector.h>
```

Public Types

- using `value_type` = T
- using `size_type` = std::size_t

Public Member Functions

- `CustomVector` () noexcept
- `CustomVector` (const `CustomVector` &other)
- `CustomVector` (`CustomVector` &&other) noexcept
- `CustomVector` & `operator=` (const `CustomVector` &other)
- `CustomVector` & `operator=` (`CustomVector` &&other) noexcept
- `~CustomVector` ()=default
- void `swap` (`CustomVector` &other) noexcept
- void `assign` (`size_type` count, const T &value)
- T * `insert` (T *pos, const T &value)
- template<typename InputIt >
T * `insert_range` (T *pos, InputIt first, InputIt last)
- template<typename InputIt >
T * `insert` (T *pos, InputIt first, InputIt last)
- template<typename... Args>
T * `emplace` (T *pos, Args &&... args)
- T * `erase` (T *pos)
- T * `erase` (T *first, T *last)
- T & `at` (`size_type` pos)
- T & `operator[]` (`size_type` pos)
- T & `front` ()
- T & `back` ()
- void `push_back` (const T &value)
- template<typename... Args>
void `emplace_back` (Args &&... args)

- void `pop_back` ()
- template<typename InputIt >
void `append_range` (InputIt first, InputIt last)
- `size_type` `size` () const noexcept
- `size_type` `capacity` () const noexcept
- `size_type` `max_size` () const noexcept
- void `reserve` (`size_type` new_cap)
- void `resize` (`size_type` count, const T &value)
- void `shrink_to_fit` ()
- void `clear` () noexcept
- bool `empty` () const noexcept
- T * `data` () noexcept
- T * `begin` () noexcept
- const T * `begin` () const noexcept
- const T * `cbegin` () const noexcept
- T * `end` () noexcept
- const T * `end` () const noexcept
- const T * `cend` () const noexcept
- std::reverse_iterator< T * > `rbegin` () noexcept
- std::reverse_iterator< const T * > `crbegin` () const noexcept
- std::reverse_iterator< T * > `rend` () noexcept
- std::reverse_iterator< const T * > `crend` () const noexcept

4.1.1 Member Typedef Documentation

4.1.1.1 `size_type`

```
template<typename T >
using CustomVector< T >::size_type = std::size_t
```

4.1.1.2 `value_type`

```
template<typename T >
using CustomVector< T >::value_type = T
```

4.1.2 Constructor & Destructor Documentation

4.1.2.1 `CustomVector()` [1/3]

```
template<typename T >
CustomVector< T >::CustomVector ( ) [noexcept]
```

4.1.2.2 `CustomVector()` [2/3]

```
template<typename T >
CustomVector< T >::CustomVector (
    const CustomVector< T > & other )
```

4.1.2.3 CustomVector() [3/3]

```
template<typename T >
CustomVector< T >::CustomVector (
    CustomVector< T > && other ) [noexcept]
```

4.1.2.4 ~CustomVector()

```
template<typename T >
CustomVector< T >::~~CustomVector ( ) [default]
```

4.1.3 Member Function Documentation

4.1.3.1 append_range()

```
template<typename T >
template<typename InputIt >
void CustomVector< T >::append_range (
    InputIt first,
    InputIt last )
```

4.1.3.2 assign()

```
template<typename T >
void CustomVector< T >::assign (
    size_type count,
    const T & value )
```

4.1.3.3 at()

```
template<typename T >
T & CustomVector< T >::at (
    size_type pos )
```

4.1.3.4 back()

```
template<typename T >
T & CustomVector< T >::back ( )
```

4.1.3.5 begin() [1/2]

```
template<typename T >
const T * CustomVector< T >::begin ( ) const [noexcept]
```

4.1.3.6 begin() [2/2]

```
template<typename T >
T * CustomVector< T >::begin ( ) [noexcept]
```

4.1.3.7 capacity()

```
template<typename T >
size_type CustomVector< T >::capacity ( ) const [noexcept]
```

4.1.3.8 cbegin()

```
template<typename T >
const T * CustomVector< T >::cbegin ( ) const [noexcept]
```

4.1.3.9 cend()

```
template<typename T >
const T * CustomVector< T >::cend ( ) const [noexcept]
```

4.1.3.10 clear()

```
template<typename T >
void CustomVector< T >::clear ( ) [noexcept]
```

4.1.3.11 crbegin()

```
template<typename T >
std::reverse_iterator< const T * > CustomVector< T >::crbegin ( ) const [noexcept]
```

4.1.3.12 crend()

```
template<typename T >
std::reverse_iterator< const T * > CustomVector< T >::crend ( ) const [noexcept]
```

4.1.3.13 data()

```
template<typename T >
T * CustomVector< T >::data ( ) [noexcept]
```

4.1.3.14 `emplace()`

```
template<typename T >
template<typename... Args>
T * CustomVector< T >::emplace (
    T * pos,
    Args &&... args )
```

4.1.3.15 `emplace_back()`

```
template<typename T >
template<typename... Args>
void CustomVector< T >::emplace_back (
    Args &&... args )
```

4.1.3.16 `empty()`

```
template<typename T >
bool CustomVector< T >::empty ( ) const [noexcept]
```

4.1.3.17 `end()` [1/2]

```
template<typename T >
const T * CustomVector< T >::end ( ) const [noexcept]
```

4.1.3.18 `end()` [2/2]

```
template<typename T >
T * CustomVector< T >::end ( ) [noexcept]
```

4.1.3.19 `erase()` [1/2]

```
template<typename T >
T * CustomVector< T >::erase (
    T * first,
    T * last )
```

4.1.3.20 `erase()` [2/2]

```
template<typename T >
T * CustomVector< T >::erase (
    T * pos )
```

4.1.3.21 `front()`

```
template<typename T >
T & CustomVector< T >::front ( )
```

4.1.3.22 insert() [1/2]

```
template<typename T >
T * CustomVector< T >::insert (
    T * pos,
    const T & value )
```

4.1.3.23 insert() [2/2]

```
template<typename T >
template<typename InputIt >
T * CustomVector< T >::insert (
    T * pos,
    InputIt first,
    InputIt last ) [inline]
```

4.1.3.24 insert_range()

```
template<typename T >
template<typename InputIt >
T * CustomVector< T >::insert_range (
    T * pos,
    InputIt first,
    InputIt last )
```

4.1.3.25 max_size()

```
template<typename T >
size_type CustomVector< T >::max_size ( ) const [noexcept]
```

4.1.3.26 operator=() [1/2]

```
template<typename T >
CustomVector & CustomVector< T >::operator= (
    const CustomVector< T > & other )
```

4.1.3.27 operator=() [2/2]

```
template<typename T >
CustomVector & CustomVector< T >::operator= (
    CustomVector< T > && other ) [noexcept]
```

4.1.3.28 operator[]()

```
template<typename T >
T & CustomVector< T >::operator[] (
    size_type pos )
```


4.1.3.29 pop_back()

```
template<typename T >
void CustomVector< T >::pop_back ( )
```

4.1.3.30 push_back()

```
template<typename T >
void CustomVector< T >::push_back (
    const T & value )
```

4.1.3.31 rbegin()

```
template<typename T >
std::reverse_iterator< T * > CustomVector< T >::rbegin ( ) [noexcept]
```

4.1.3.32 rend()

```
template<typename T >
std::reverse_iterator< T * > CustomVector< T >::rend ( ) [noexcept]
```

4.1.3.33 reserve()

```
template<typename T >
void CustomVector< T >::reserve (
    size_type new_cap )
```

4.1.3.34 resize()

```
template<typename T >
void CustomVector< T >::resize (
    size_type count,
    const T & value )
```

4.1.3.35 shrink_to_fit()

```
template<typename T >
void CustomVector< T >::shrink_to_fit ( )
```

4.1.3.36 size()

```
template<typename T >
size_type CustomVector< T >::size ( ) const [noexcept]
```

4.1.3.37 swap()

```
template<typename T >
void CustomVector< T >::swap (
    CustomVector< T > & other ) [noexcept]
```

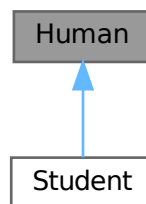
The documentation for this class was generated from the following file:

- include/custom-vector.h

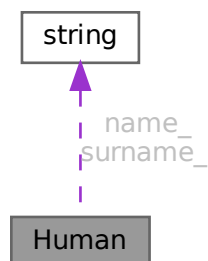
4.2 Human Class Reference

```
#include <human.h>
```

Inheritance diagram for Human:



Collaboration diagram for Human:



Public Member Functions

- `Human()`=default
- `Human(string name, string surname)`
- `Human(const Human &)=default`
- `Human(Human &&) noexcept=default`
- `Human & operator=(const Human &)=default`
- `Human & operator=(Human &&) noexcept=default`
- `virtual ~Human()`=0
- `const string & name()` const
- `const string & surname()` const
- `void setName(string name)`
- `void setSurname(string surname)`

Protected Attributes

- `string name_`
- `string surname_`

4.2.1 Constructor & Destructor Documentation

4.2.1.1 Human() [1/4]

```
Human::Human ( ) [default]
```

4.2.1.2 Human() [2/4]

```
Human::Human (
    string name,
    string surname ) [inline]
```

4.2.1.3 Human() [3/4]

```
Human::Human (
    const Human & ) [default]
```

4.2.1.4 Human() [4/4]

```
Human::Human (
    Human && ) [default], [noexcept]
```

4.2.1.5 ~Human()

```
Human::~~Human ( ) [inline], [pure virtual], [default]
```

4.2.2 Member Function Documentation

4.2.2.1 name()

```
const string & Human::name ( ) const [inline]
```

4.2.2.2 operator=() [1/2]

```
Human & Human::operator= (
    const Human & ) [default]
```

4.2.2.3 operator=() [2/2]

```
Human & Human::operator= (
    Human && ) [default], [noexcept]
```

4.2.2.4 setName()

```
void Human::setName (
    string name ) [inline]
```

4.2.2.5 setSurname()

```
void Human::setSurname (
    string surname ) [inline]
```

4.2.2.6 surname()

```
const string & Human::surname ( ) const [inline]
```

4.2.3 Member Data Documentation

4.2.3.1 name_

```
string Human::name_ [protected]
```

4.2.3.2 surname_

```
string Human::surname_ [protected]
```

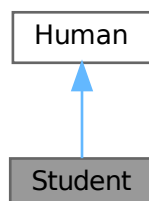
The documentation for this class was generated from the following file:

- [include/human.h](#)

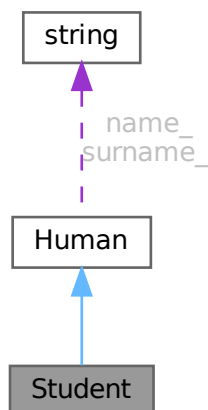
4.3 Student Class Reference

```
#include <student.h>
```

Inheritance diagram for Student:



Collaboration diagram for Student:



Public Member Functions

- [Student](#) ()
- [Student](#) (string [name](#), string [surname](#))
- [Student](#) (string [name](#), string [surname](#), vector< int > [homeworkGrades](#), int [examGrade](#))
- [Student](#) (std::istream &is)
- [Student](#) (const [Student](#) &other)
- [Student](#) ([Student](#) &&other) noexcept
- [Student](#) & [operator=](#) (const [Student](#) &other)
- [Student](#) & [operator=](#) ([Student](#) &&other) noexcept
- [~Student](#) ()

- int [examGrade](#) () const
- const vector< int > & [homeworkGrades](#) () const
- double [finalGradeAverage](#) () const
- double [finalGradeMedian](#) () const
- void [setExamGrade](#) (int [examGrade](#))
- void [setHomeworkGrades](#) (vector< int > [homeworkGrades](#))
- std::istream & [readStudent](#) (std::istream &is)
- const string & [name](#) () const
- const string & [surname](#) () const
- void [setName](#) (string [name](#))
- void [setSurname](#) (string [surname](#))

Public Member Functions inherited from [Human](#)

- [Human](#) ()=default
- [Human](#) (string [name](#), string [surname](#))
- [Human](#) (const [Human](#) &)=default
- [Human](#) ([Human](#) &&) noexcept=default
- [Human](#) & [operator=](#) (const [Human](#) &)=default
- [Human](#) & [operator=](#) ([Human](#) &&) noexcept=default
- virtual [~Human](#) ()=0
- const string & [name](#) () const
- const string & [surname](#) () const
- void [setName](#) (string [name](#))
- void [setSurname](#) (string [surname](#))

Friends

- std::istream & [operator>>](#) (std::istream &is, [Student](#) &student)
- std::ostream & [operator<<](#) (std::ostream &os, const [Student](#) &student)

Additional Inherited Members

Protected Attributes inherited from [Human](#)

- string [name_](#)
- string [surname_](#)

4.3.1 Constructor & Destructor Documentation

4.3.1.1 [Student\(\)](#) [1/6]

```
Student::Student ( ) [inline]
```

4.3.1.2 [Student\(\)](#) [2/6]

```
Student::Student (
    string name,
    string surname ) [inline]
```

4.3.1.3 Student() [3/6]

```
Student::Student (
    string name,
    string surname,
    vector< int > homeworkGrades,
    int examGrade ) [inline]
```

4.3.1.4 Student() [4/6]

```
Student::Student (
    std::istream & is )
```

4.3.1.5 Student() [5/6]

```
Student::Student (
    const Student & other ) [inline]
```

4.3.1.6 Student() [6/6]

```
Student::Student (
    Student && other ) [inline], [noexcept]
```

4.3.1.7 ~Student()

```
Student::~~Student ( ) [inline]
```

4.3.2 Member Function Documentation

4.3.2.1 examGrade()

```
int Student::examGrade ( ) const [inline]
```

4.3.2.2 finalGradeAverage()

```
double Student::finalGradeAverage ( ) const
```

4.3.2.3 finalGradeMedian()

```
double Student::finalGradeMedian ( ) const
```

4.3.2.4 homeworkGrades()

```
const vector< int > & Student::homeworkGrades ( ) const [inline]
```

4.3.2.5 name()

```
const string & Human::name ( ) const [inline]
```

4.3.2.6 operator=() [1/2]

```
Student & Student::operator= (
    const Student & other ) [inline]
```

4.3.2.7 operator=() [2/2]

```
Student & Student::operator= (
    Student && other ) [inline], [noexcept]
```

4.3.2.8 readStudent()

```
std::istream & Student::readStudent (
    std::istream & is )
```

4.3.2.9 setExamGrade()

```
void Student::setExamGrade (
    int examGrade ) [inline]
```

4.3.2.10 setHomeworkGrades()

```
void Student::setHomeworkGrades (
    vector< int > homeworkGrades ) [inline]
```

4.3.2.11 setName()

```
void Human::setName (
    string name ) [inline]
```

4.3.2.12 setSurname()

```
void Human::setSurname (
    string surname ) [inline]
```

4.3.2.13 surname()

```
const string & Human::surname ( ) const [inline]
```


4.3.3 Friends And Related Symbol Documentation

4.3.3.1 `operator<<`

```
std::ostream & operator<< (  
    std::ostream & os,  
    const Student & student ) [friend]
```

4.3.3.2 `operator>>`

```
std::istream & operator>> (  
    std::istream & is,  
    Student & student ) [friend]
```

The documentation for this class was generated from the following files:

- [include/student.h](#)
- [src/student.cpp](#)

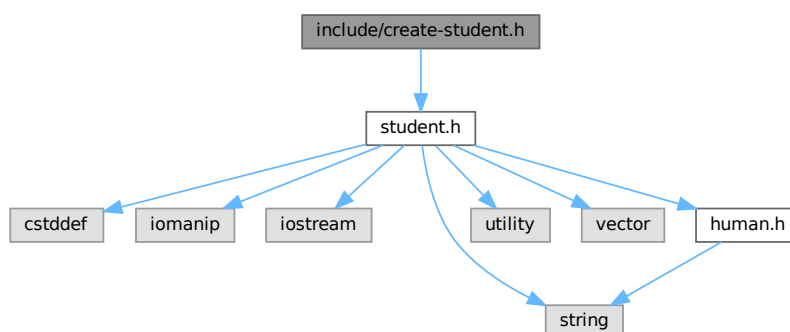
Chapter 5

File Documentation

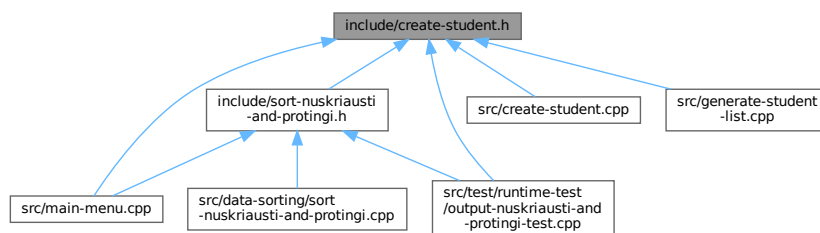
5.1 include/create-student.h File Reference

```
#include "student.h"
```

Include dependency graph for create-student.h:



This graph shows which files directly or indirectly include this file:



Functions

- string [randomName](#) ()
- string [randomSurname](#) ()
- [Student](#) [createStudentManual](#) ()
- [Student](#) [createStudentRandomGrades](#) ()
- [Student](#) [createStudentFullyRandom](#) ()
- `template<typename T >`
`T` [createStudentsFromFile](#) (const string &filename)

5.1.1 Function Documentation

5.1.1.1 [createStudentFullyRandom\(\)](#)

```
Student createStudentFullyRandom ( )
```

5.1.1.2 [createStudentManual\(\)](#)

```
Student createStudentManual ( )
```

5.1.1.3 [createStudentRandomGrades\(\)](#)

```
Student createStudentRandomGrades ( )
```

5.1.1.4 [createStudentsFromFile\(\)](#)

```
template<typename T >
T createStudentsFromFile (
    const string & filename )
```

5.1.1.5 [randomName\(\)](#)

```
string randomName ( )
```

5.1.1.6 [randomSurname\(\)](#)

```
string randomSurname ( )
```

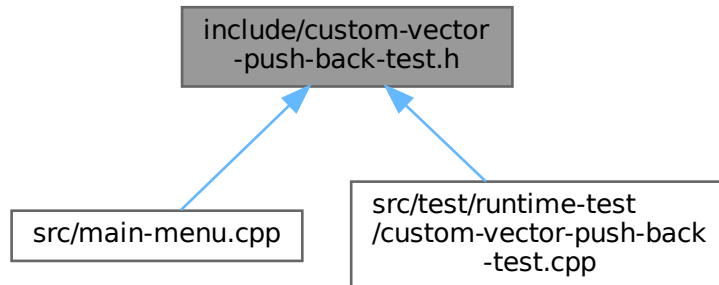
5.2 [create-student.h](#)

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "student.h"
00004
00005 string randomName();
00006 string randomSurname();
00007
00008 Student createStudentManual();
00009 Student createStudentRandomGrades();
00010 Student createStudentFullyRandom();
00011
00012 template <typename T>
00013 T createStudentsFromFile(const string& filename);
```

5.3 include/custom-vector-push-back-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [handleCustomVectorPushBackTest](#) ()

5.3.1 Function Documentation

5.3.1.1 handleCustomVectorPushBackTest()

```
void handleCustomVectorPushBackTest ( )
```

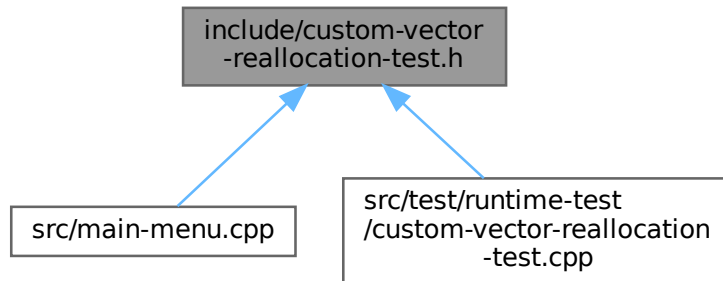
5.4 custom-vector-push-back-test.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void handleCustomVectorPushBackTest ();
```

5.5 include/custom-vector-reallocation-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [handleCustomVectorReallocationTest](#) ()

5.5.1 Function Documentation

5.5.1.1 handleCustomVectorReallocationTest()

```
void handleCustomVectorReallocationTest ( )
```

5.6 custom-vector-reallocation-test.h

[Go to the documentation of this file.](#)

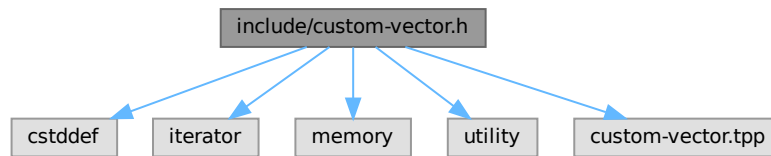
```
00001 #pragma once
00002
00003 void handleCustomVectorReallocationTest ();
```

5.7 include/custom-vector.h File Reference

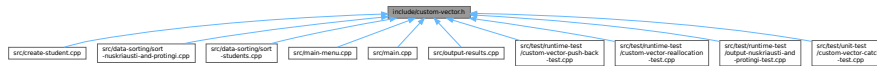
```
#include <cstddef>
#include <iterator>
#include <memory>
#include <utility>
```

```
#include "custom-vector.hpp"
```

Include dependency graph for custom-vector.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [CustomVector< T >](#)

Functions

- `template<typename T >`
`bool operator== (const CustomVector< T > &lhs, const CustomVector< T > &rhs)`
- `template<typename T >`
`bool operator!= (const CustomVector< T > &lhs, const CustomVector< T > &rhs)`
- `template<typename T >`
`bool operator< (const CustomVector< T > &lhs, const CustomVector< T > &rhs)`
- `template<typename T >`
`bool operator<= (const CustomVector< T > &lhs, const CustomVector< T > &rhs)`
- `template<typename T >`
`bool operator> (const CustomVector< T > &lhs, const CustomVector< T > &rhs)`
- `template<typename T >`
`bool operator>= (const CustomVector< T > &lhs, const CustomVector< T > &rhs)`

5.7.1 Function Documentation

5.7.1.1 `operator"!=()`

```

template<typename T >
bool operator!= (
    const CustomVector< T > & lhs,
    const CustomVector< T > & rhs )
  
```

5.7.1.2 operator<()

```
template<typename T >
bool operator< (
    const CustomVector< T > & lhs,
    const CustomVector< T > & rhs )
```

5.7.1.3 operator<=()

```
template<typename T >
bool operator<= (
    const CustomVector< T > & lhs,
    const CustomVector< T > & rhs )
```

5.7.1.4 operator==()

```
template<typename T >
bool operator== (
    const CustomVector< T > & lhs,
    const CustomVector< T > & rhs )
```

5.7.1.5 operator>()

```
template<typename T >
bool operator> (
    const CustomVector< T > & lhs,
    const CustomVector< T > & rhs )
```

5.7.1.6 operator>=()

```
template<typename T >
bool operator>= (
    const CustomVector< T > & lhs,
    const CustomVector< T > & rhs )
```

5.8 custom-vector.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <cstdint>
00004 #include <iterator>
00005 #include <memory>
00006 #include <utility>
00007
00008 template <typename T>
00009 class CustomVector {
00010 public:
00011     using value_type = T;
00012     using size_type = std::size_t;
00013
00014     CustomVector() noexcept;
00015     CustomVector(const CustomVector& other);
```



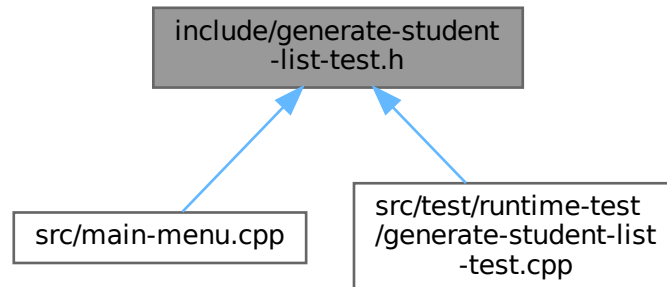
```

00016     CustomVector(CustomVector&& other) noexcept;
00017     CustomVector& operator=(const CustomVector& other);
00018     CustomVector& operator=(CustomVector&& other) noexcept;
00019     ~CustomVector() = default;
00020
00021     void swap(CustomVector& other) noexcept;
00022
00023     void assign(size_type count, const T& value);
00024     T* insert(T* pos, const T& value);
00025
00026     template <typename InputIt>
00027     T* insert_range(T* pos, InputIt first, InputIt last);
00028
00029     template <typename InputIt>
00030     T* insert(T* pos, InputIt first, InputIt last) {
00031         return insert_range(pos, first, last);
00032     }
00033
00034     template <typename... Args>
00035     T* emplace(T* pos, Args&&... args);
00036
00037     T* erase(T* pos);
00038     T* erase(T* first, T* last);
00039
00040     T& at(size_type pos);
00041     T& operator[](size_type pos);
00042     T& front();
00043     T& back();
00044     void push_back(const T& value);
00045
00046     template <typename... Args>
00047     void emplace_back(Args&&... args);
00048
00049     void pop_back();
00050
00051     template <typename InputIt>
00052     void append_range(InputIt first, InputIt last);
00053
00054     size_type size() const noexcept;
00055     size_type capacity() const noexcept;
00056     size_type max_size() const noexcept;
00057     void reserve(size_type new_cap);
00058     void resize(size_type count, const T& value);
00059     void shrink_to_fit();
00060     void clear() noexcept;
00061     bool empty() const noexcept;
00062     T* data() noexcept;
00063     T* begin() noexcept;
00064     const T* begin() const noexcept;
00065     const T* cbegin() const noexcept;
00066     T* end() noexcept;
00067     const T* end() const noexcept;
00068     const T* cend() const noexcept;
00069     std::reverse_iterator<T*> rbegin() noexcept;
00070     std::reverse_iterator<const T*> crbegin() const noexcept;
00071     std::reverse_iterator<T*> rend() noexcept;
00072     std::reverse_iterator<const T*> crend() const noexcept;
00073
00074 private:
00075     void reset() noexcept;
00076
00077     std::unique_ptr<T[]> data_;
00078     size_type size_;
00079     size_type capacity_;
00080 };
00081
00082 template <typename T>
00083 bool operator==(const CustomVector<T>& lhs, const CustomVector<T>& rhs);
00084
00085 template <typename T>
00086 bool operator!=(const CustomVector<T>& lhs, const CustomVector<T>& rhs);
00087
00088 template <typename T>
00089 bool operator<(const CustomVector<T>& lhs, const CustomVector<T>& rhs);
00090
00091 template <typename T>
00092 bool operator<=(const CustomVector<T>& lhs, const CustomVector<T>& rhs);
00093
00094 template <typename T>
00095 bool operator>(const CustomVector<T>& lhs, const CustomVector<T>& rhs);
00096
00097 template <typename T>
00098 bool operator>=(const CustomVector<T>& lhs, const CustomVector<T>& rhs);
00099
00100 #include "custom-vector.tpp"

```

5.9 include/generate-student-list-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [handleGenerateStudentListTest](#) ()

5.9.1 Function Documentation

5.9.1.1 handleGenerateStudentListTest()

```
void handleGenerateStudentListTest ( )
```

5.10 generate-student-list-test.h

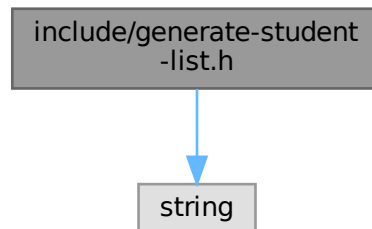
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void handleGenerateStudentListTest ();
```

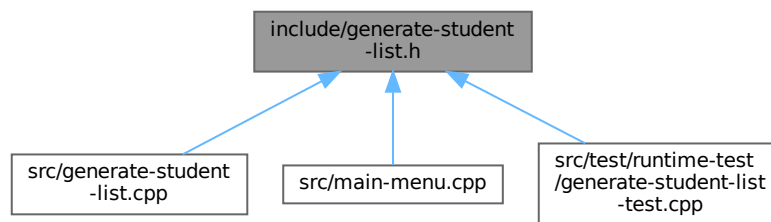
5.11 include/generate-student-list.h File Reference

```
#include <string>
```

Include dependency graph for generate-student-list.h:



This graph shows which files directly or indirectly include this file:



Functions

- void [generateStudentListFile](#) (int studentCount, const std::string &filename)
- void [handleGenerateStudentListFile](#) ()

5.11.1 Function Documentation

5.11.1.1 generateStudentListFile()

```
void generateStudentListFile (
    int studentCount,
    const std::string & filename )
```

5.11.1.2 handleGenerateStudentListFile()

```
void handleGenerateStudentListFile ( )
```

5.12 generate-student-list.h

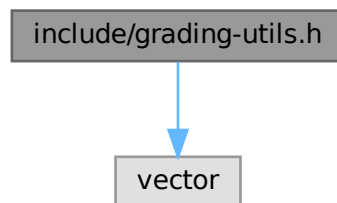
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <string>
00004
00005 void generateStudentListFile(int studentCount, const std::string& filename);
00006 void handleGenerateStudentListFile();
```

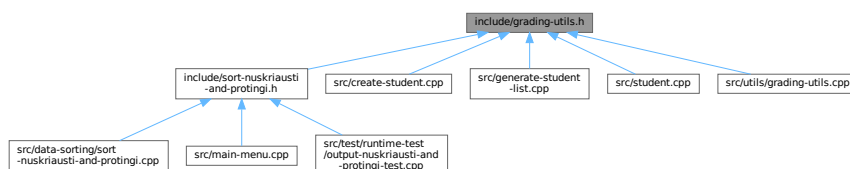
5.13 include/grading-utils.h File Reference

```
#include <vector>
```

Include dependency graph for grading-utils.h:



This graph shows which files directly or indirectly include this file:



Functions

- int [randomGrade](#) (int min=1, int max=10)
- double [calculateFinalGradeAverage](#) (const vector< int > &homeworkGrades, int examGrade)
- double [calculateFinalGradeMedian](#) (const vector< int > &homeworkGrades, int examGrade)

5.13.1 Function Documentation

5.13.1.1 calculateFinalGradeAverage()

```
double calculateFinalGradeAverage (
    const vector< int > & homeworkGrades,
    int examGrade )
```

5.13.1.2 calculateFinalGradeMedian()

```
double calculateFinalGradeMedian (
    const vector< int > & homeworkGrades,
    int examGrade )
```

5.13.1.3 randomGrade()

```
int randomGrade (
    int min = 1,
    int max = 10 )
```

5.14 grading-utils.h

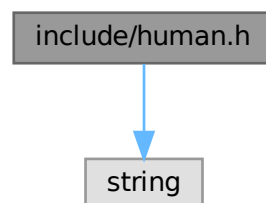
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <vector>
00004
00005 using std::vector;
00006
00007 int randomGrade(int min = 1, int max = 10);
00008 double calculateFinalGradeAverage(const vector<int>& homeworkGrades, int examGrade);
00009 double calculateFinalGradeMedian(const vector<int>& homeworkGrades, int examGrade);
```

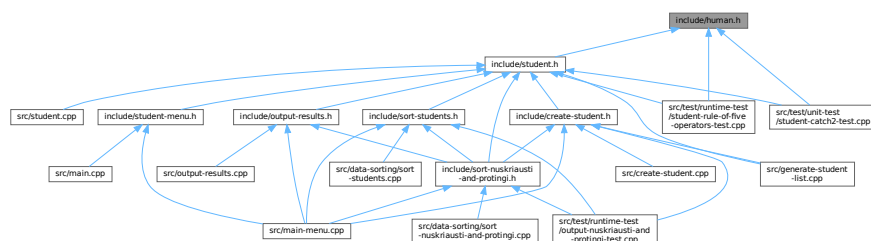
5.15 include/human.h File Reference

```
#include <string>
```

Include dependency graph for human.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Human](#)

5.16 human.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include <string>
00004
00005 using std::string;
00006
00007 class Human {
00008 protected:
00009     string name_;
00010     string surname_;
00011
00012 public:
00013     Human() = default;
00014     Human(string name, string surname) : name_(name), surname_(surname) {}
00015     Human(const Human&) = default;
00016     Human(Human&&) noexcept = default;
00017     Human& operator=(const Human&) = default;
00018     Human& operator=(Human&&) noexcept = default;
00019     virtual ~Human() = 0;
00020
00021     inline const string& name() const { return name_; }
00022     inline const string& surname() const { return surname_; }
00023
00024     inline void setName(string name) { name_ = name; }
00025     inline void setSurname(string surname) { surname_ = surname; }
00026 };
00027
00028 inline Human::~Human() = default;

```

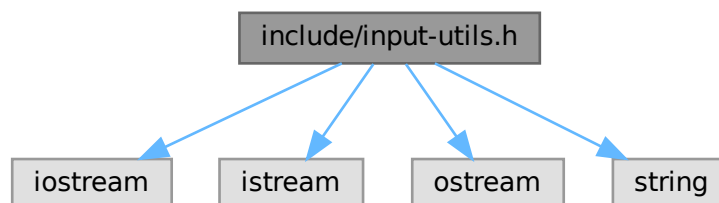
5.17 include/input-utils.h File Reference

```

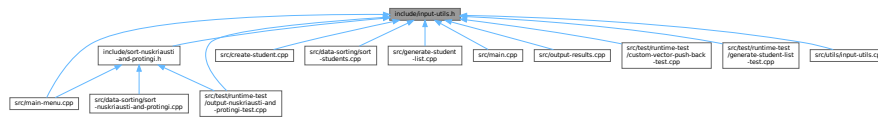
#include <iostream>
#include <istream>
#include <ostream>
#include <string>

```

Include dependency graph for input-utils.h:



This graph shows which files directly or indirectly include this file:



Functions

- int [readIntInRange](#) (const string &prompt, int min, int max)
- string [readSingleStringToken](#) (const string &prompt)
- char [readYesOrNo](#) (const string &prompt)
- int [readSingleIntToken](#) (const string &prompt)

5.17.1 Function Documentation

5.17.1.1 readIntInRange()

```
int readIntInRange (
    const string & prompt,
    int min,
    int max )
```

5.17.1.2 readSingleIntToken()

```
int readSingleIntToken (
    const string & prompt )
```

5.17.1.3 readSingleStringToken()

```
string readSingleStringToken (
    const string & prompt )
```

5.17.1.4 readYesOrNo()

```
char readYesOrNo (
    const string & prompt )
```

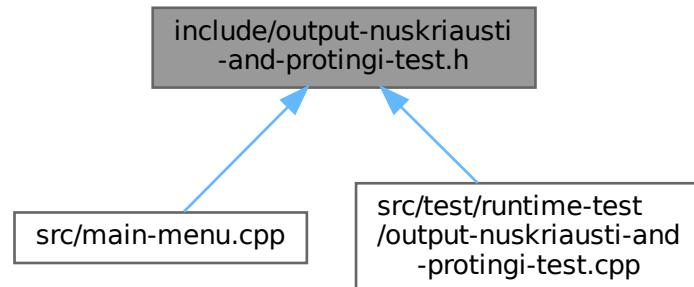
5.18 input-utils.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include <iostream>
00004 #include <istream>
00005 #include <ostream>
00006 #include <string>
00007
00008 using std::istream;
00009 using std::ostream;
00010 using std::string;
00011
00012 int readIntInRange(const string& prompt, int min, int max);
00013 string readSingleStringToken(const string& prompt);
00014 char readYesOrNo(const string& prompt);
00015 int readSingleIntToken(const string& prompt);
```

5.19 include/output-nuskriausti-and-protingi-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles\(\)](#)

5.19.1 Function Documentation

5.19.1.1 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles()

```
void outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles ( )
```

5.20 output-nuskriausti-and-protingi-test.h

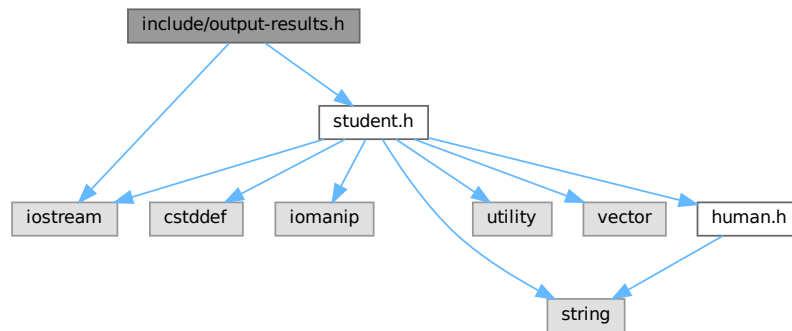
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles();
```

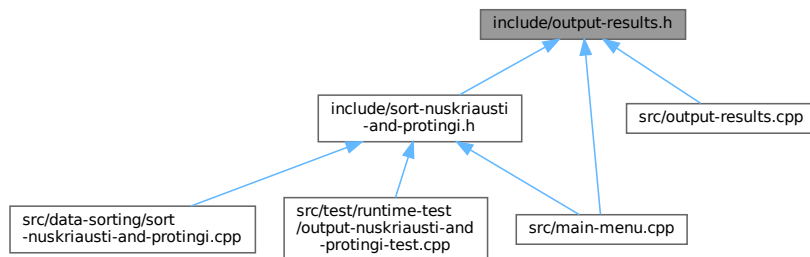
5.21 include/output-results.h File Reference

```
#include "student.h"
#include <iostream>
```


Include dependency graph for output-results.h:



This graph shows which files directly or indirectly include this file:



Functions

- `template<typename T >`
void `outputResults` (const T &students, std::ostream &out)
- `template<typename T >`
void `chooseOutputAndPrint` (T &students)

5.21.1 Function Documentation

5.21.1.1 chooseOutputAndPrint()

```

template<typename T >
void chooseOutputAndPrint (
    T & students )

```

5.21.1.2 outputResults()

```

template<typename T >
void outputResults (
    const T & students,
    std::ostream & out )

```

5.22 output-results.h

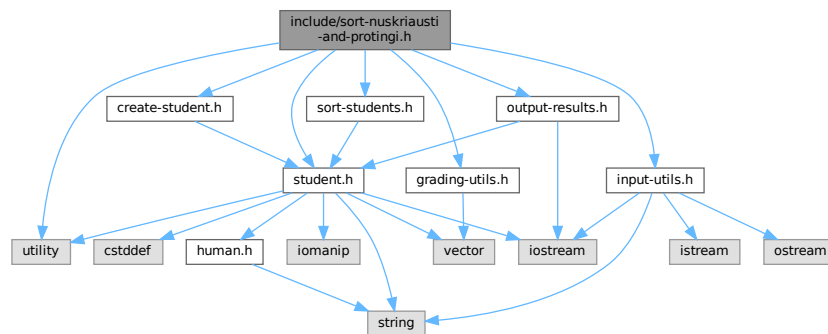
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "student.h"
00004
00005 #include <iostream>
00006
00007 template <typename T>
00008 void outputResults(const T& students, std::ostream& out);
00009
00010 template <typename T>
00011 void chooseOutputAndPrint(T& students);
```

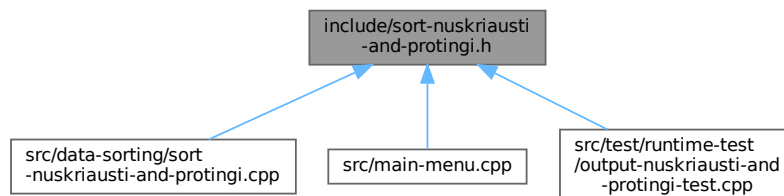
5.23 include/sort-nuskriausti-and-protingi.h File Reference

```
#include "create-student.h"
#include "input-utils.h"
#include "output-results.h"
#include "sort-students.h"
#include "student.h"
#include "grading-utils.h"
#include <utility>
```

Include dependency graph for sort-nuskriausti-and-protingi.h:



This graph shows which files directly or indirectly include this file:



Functions

- `template<typename T >`
`void handleSortNuskriaustiAndProtingiFromFile ( )`
- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (const T &students)`
- `template<typename T >`
`T sortNuskriaustiAndProtingiSecondStrategy (T &students)`
- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (T &students)`

5.23.1 Function Documentation

5.23.1.1 [handleSortNuskriaustiAndProtingiFromFile\(\)](#)

```
template<typename T >
void handleSortNuskriaustiAndProtingiFromFile ( )
```

5.23.1.2 [sortNuskriaustiAndProtingiFirstStrategy\(\)](#)

```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (
    const T & students )
```

5.23.1.3 [sortNuskriaustiAndProtingiSecondStrategy\(\)](#)

```
template<typename T >
T sortNuskriaustiAndProtingiSecondStrategy (
    T & students )
```

5.23.1.4 [sortNuskriaustiAndProtingiThirdStrategy\(\)](#)

```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (
    T & students )
```

5.24 sort-nuskriausti-and-protingi.h

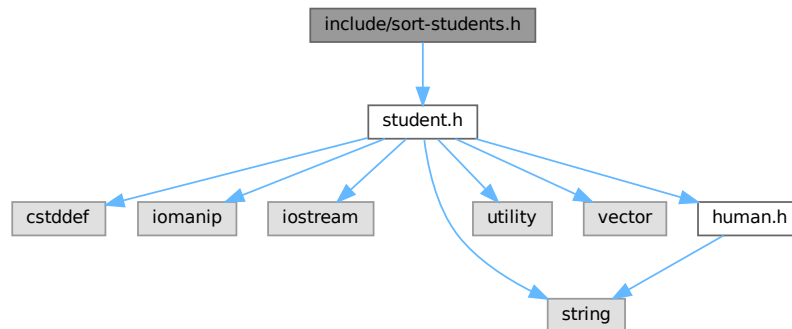
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "create-student.h"
00004 #include "input-utils.h"
00005 #include "output-results.h"
00006 #include "sort-students.h"
00007 #include "student.h"
00008 #include "grading-utils.h"
00009
00010 #include <utility>
00011
00012 template <typename T>
00013 void handleSortNuskriaustiAndProtingiFromFile();
00014
00015 template <typename T>
00016 std::pair<T, T> sortNuskriaustiAndProtingiFirstStrategy(const T& students);
00017
00018 template <typename T>
00019 T sortNuskriaustiAndProtingiSecondStrategy(T& students);
00020
00021 template <typename T>
00022 std::pair<T, T> sortNuskriaustiAndProtingiThirdStrategy(T& students);
```

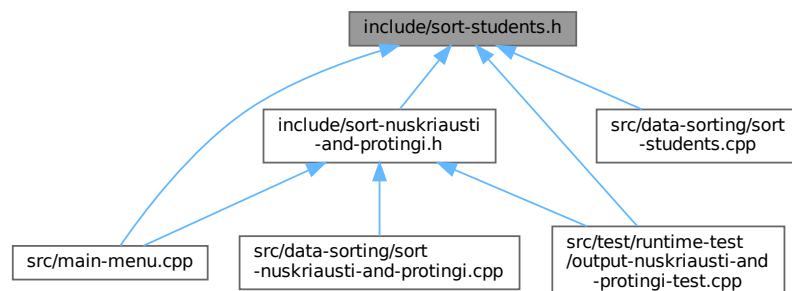
5.25 include/sort-students.h File Reference

```
#include "student.h"
```

Include dependency graph for sort-students.h:



This graph shows which files directly or indirectly include this file:



Functions

- `template<typename T >`
void `runSortStudentsChoicePrompt` (T &students)
- `template<typename T >`
void `sortStudents` (T &students, int sortChoice, bool ascending)

5.25.1 Function Documentation

5.25.1.1 runSortStudentsChoicePrompt()

```
template<typename T >
void runSortStudentsChoicePrompt (
    T & students )
```

5.25.1.2 sortStudents()

```
template<typename T >
void sortStudents (
    T & students,
    int sortChoice,
    bool ascending )
```

5.26 sort-students.h

[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 #include "student.h"
00004
00005 template <typename T>
00006 void runSortStudentsChoicePrompt (T& students);
00007
00008 template <typename T>
00009 void sortStudents (T& students, int sortChoice, bool ascending);
```

5.27 include/student-app.h File Reference

Functions

- void [runStudentApp](#) ()

5.27.1 Function Documentation

5.27.1.1 runStudentApp()

```
void runStudentApp ( )
```

5.28 student-app.h

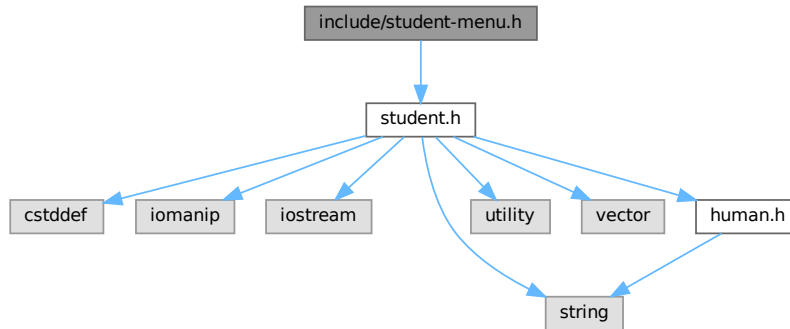
[Go to the documentation of this file.](#)

```
00001 #pragma once
00002
00003 void runStudentApp ();
```

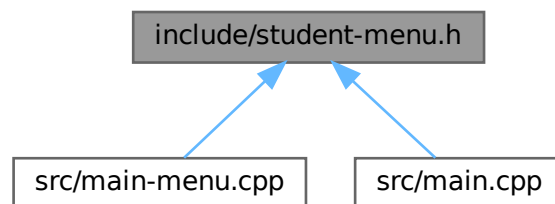
5.29 include/student-menu.h File Reference

```
#include "student.h"
```

Include dependency graph for student-menu.h:



This graph shows which files directly or indirectly include this file:



Functions

- void [showMainMenu](#) ()
- template<typename T >
void [handleMenuChoice](#) (int choice, T &students)
- template<typename T >
void [startMainMenu](#) ()

5.29.1 Function Documentation

5.29.1.1 handleMenuChoice()

```
template<typename T >
void handleMenuChoice (
    int choice,
    T & students )
```

5.29.1.2 showMainMenu()

```
void showMainMenu ( )
```

5.29.1.3 startMainMenu()

```
template<typename T >  
void startMainMenu ( )
```

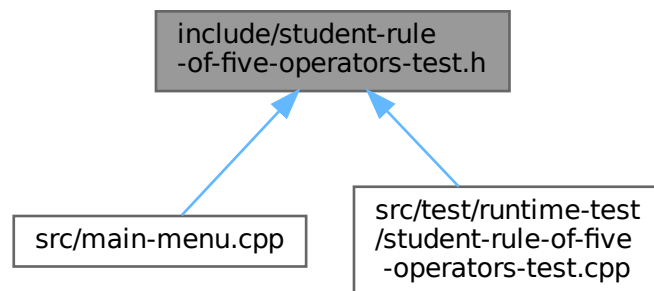
5.30 student-menu.h

[Go to the documentation of this file.](#)

```
00001 #pragma once  
00002  
00003 #include "student.h"  
00004  
00005 void showMainMenu();  
00006  
00007 template <typename T>  
00008 void handleMenuChoice(int choice, T& students);  
00009  
00010 template <typename T>  
00011 void startMainMenu();
```

5.31 include/student-rule-of-five-operators-test.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- void [handleStudentRuleOfFiveAndOperatorsTest](#) ()

5.34 student.h

[Go to the documentation of this file.](#)

```

00001 #pragma once
00002
00003 #include <cstdint>
00004 #include <iomanip>
00005 #include <iostream>
00006 #include <string>
00007 #include <utility>
00008 #include <vector>
00009
00010 #include "human.h"
00011
00012 using std::string;
00013 using std::vector;
00014 using std::cout;
00015 using std::cin;
00016 using std::endl;
00017 using std::setw;
00018 using std::left;
00019 using std::right;
00020
00021 class Student : public Human {
00022 private:
00023     int examGrade_;
00024     vector<int> homeworkGrades_;
00025
00026 public:
00027     Student() : Human(), examGrade_(0) {}
00028
00029     Student(string name, string surname) : Human(name, surname), examGrade_(0) {}
00030     Student(string name, string surname, vector<int> homeworkGrades, int examGrade) : Human(name,
00031 surname), examGrade_(examGrade), homeworkGrades_(homeworkGrades) {}
00032     Student(std::istream& is);
00033     Student(const Student& other)
00034         : Human(other),
00035           examGrade_(other.examGrade_),
00036           homeworkGrades_(other.homeworkGrades_) {}
00037
00038     Student(Student&& other) noexcept
00039         : Human(std::move(other)),
00040           examGrade_(other.examGrade_),
00041           homeworkGrades_(std::move(other.homeworkGrades_)) {
00042         other.examGrade_ = 0;
00043     }
00044     Student& operator=(const Student& other) {
00045         if (this == &other) {
00046             return *this;
00047         }
00048         Human::operator=(other);
00049         examGrade_ = other.examGrade_;
00050         homeworkGrades_ = other.homeworkGrades_;
00051
00052         return *this;
00053     }
00054     Student& operator=(Student&& other) noexcept {
00055         if (this == &other) {
00056             return *this;
00057         }
00058         Human::operator=(std::move(other));
00059         examGrade_ = other.examGrade_;
00060         homeworkGrades_ = std::move(other.homeworkGrades_);
00061         other.examGrade_ = 0;
00062
00063         return *this;
00064     }
00065
00066     ~Student() {
00067         name_.clear();
00068         surname_.clear();
00069         homeworkGrades_.clear();
00070         examGrade_ = 0;
00071     }
00072
00073     using Human::name;
00074     using Human::surname;
00075     inline int examGrade() const { return examGrade_; }
00076     inline const vector<int>& homeworkGrades() const { return homeworkGrades_; }
00077     double finalGradeAverage() const;
00078     double finalGradeMedian() const;
00079
00080     using Human::setName;
00081     using Human::setSurname;

```

```

00082     inline void setExamGrade(int examGrade) { examGrade_ = examGrade; }
00083     inline void setHomeworkGrades(vector<int> homeworkGrades) { homeworkGrades_ = homeworkGrades; }
00084
00085     std::istream& readStudent(std::istream& is);
00086     friend std::istream& operator>>(std::istream& is, Student& student);
00087     friend std::ostream& operator<<(std::ostream& os, const Student& student);
00088 };

```

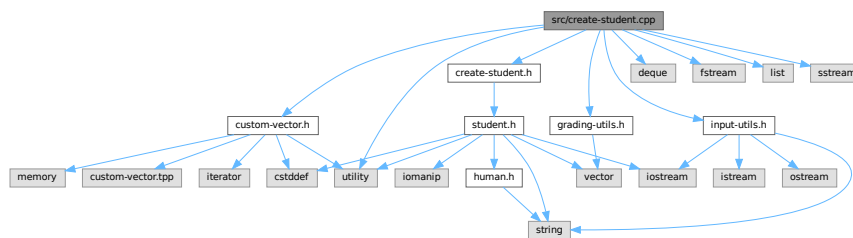
5.35 src/create-student.cpp File Reference

```

#include "create-student.h"
#include "custom-vector.h"
#include "input-utils.h"
#include "grading-utils.h"
#include <deque>
#include <fstream>
#include <list>
#include <sstream>
#include <utility>

```

Include dependency graph for create-student.cpp:



Functions

- [Student createStudentManual\(\)](#)
- [Student createStudentRandomGrades\(\)](#)
- [string randomName\(\)](#)
- [string randomSurname\(\)](#)
- [Student createStudentFullyRandom\(\)](#)
- [template<typename T> T createStudentsFromFile\(const string &filename\)](#)
- [template vector< Student > createStudentsFromFile\(const string &filename\)](#)

5.35.1 Function Documentation

5.35.1.1 createStudentFullyRandom()

```
Student createStudentFullyRandom ( )
```

5.35.1.2 createStudentManual()

```
Student createStudentManual ( )
```

5.35.1.3 createStudentRandomGrades()

```
Student createStudentRandomGrades ( )
```

5.35.1.4 createStudentsFromFile() [1/2]

```
template<typename T >
template CustomVector< Student > createStudentsFromFile (
    const string & filename )
```

5.35.1.5 createStudentsFromFile() [2/2]

```
template vector< Student > createStudentsFromFile (
    const string & filename )
```

5.35.1.6 randomName()

```
string randomName ( )
```

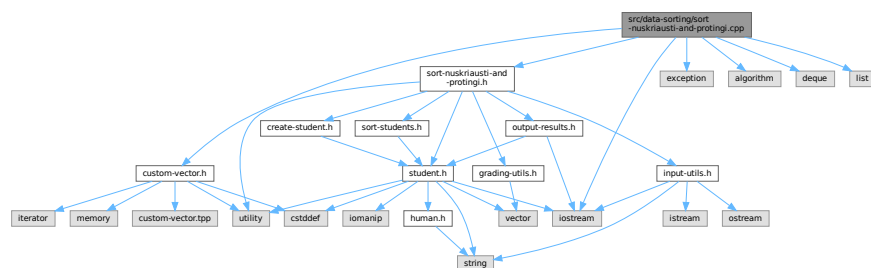
5.35.1.7 randomSurname()

```
string randomSurname ( )
```

5.36 src/data-sorting/sort-nuskriausti-and-protingi.cpp File Reference

```
#include "sort-nuskriausti-and-protingi.h"
#include "custom-vector.h"
#include <exception>
#include <iostream>
#include <algorithm>
#include <deque>
#include <list>
```

Include dependency graph for sort-nuskriausti-and-protingi.cpp:



Functions

- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (const T &students)`
- `template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiFirstStrategy (const vector< Student > &)`
- `template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiFirstStrategy (const std::list< Student > &)`
- `template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiFirstStrategy (const std::deque< Student > &)`
- `template std::pair< CustomVector< Student >, CustomVector< Student > > sortNuskriaustiAndProtingiFirstStrategy (const CustomVector< Student > &)`
- `template<typename T >`
`T sortNuskriaustiAndProtingiSecondStrategy (T &students)`
- `template vector< Student > sortNuskriaustiAndProtingiSecondStrategy (vector< Student > &)`
- `template std::list< Student > sortNuskriaustiAndProtingiSecondStrategy (std::list< Student > &)`
- `template std::deque< Student > sortNuskriaustiAndProtingiSecondStrategy (std::deque< Student > &)`
- `template CustomVector< Student > sortNuskriaustiAndProtingiSecondStrategy (CustomVector< Student > &)`
- `template<typename T >`
`std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (T &students)`
- `template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiThirdStrategy (vector< Student > &)`
- `template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiThirdStrategy (std::list< Student > &)`
- `template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiThirdStrategy (std::deque< Student > &)`
- `template std::pair< CustomVector< Student >, CustomVector< Student > > sortNuskriaustiAndProtingiThirdStrategy (CustomVector< Student > &)`
- `template<typename T >`
`void handleSortNuskriaustiAndProtingiFromFile ()`
- `template void handleSortNuskriaustiAndProtingiFromFile< vector< Student > > ()`
- `template void handleSortNuskriaustiAndProtingiFromFile< std::list< Student > > ()`
- `template void handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > > ()`
- `template void handleSortNuskriaustiAndProtingiFromFile< CustomVector< Student > > ()`

5.36.1 Function Documentation

5.36.1.1 handleSortNuskriaustiAndProtingiFromFile()

```
template<typename T >
void handleSortNuskriaustiAndProtingiFromFile ( )
```

5.36.1.2 handleSortNuskriaustiAndProtingiFromFile< CustomVector< Student > >()

```
template void handleSortNuskriaustiAndProtingiFromFile< CustomVector< Student > > ( )
```

5.36.1.3 handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > >()

```
template void handleSortNuskriaustiAndProtingiFromFile< std::deque< Student > > ( )
```

5.36.1.4 handleSortNuskriaustiAndProtingiFromFile< std::list< Student > >()

```
template void handleSortNuskriaustiAndProtingiFromFile< std::list< Student > > ( )
```

5.36.1.5 handleSortNuskriaustiAndProtingiFromFile< vector< Student > >()

```
template void handleSortNuskriaustiAndProtingiFromFile< vector< Student > > ( )
```

5.36.1.6 sortNuskriaustiAndProtingiFirstStrategy() [1/5]

```
template std::pair< CustomVector< Student >, CustomVector< Student > > sortNuskriaustiAndProtingiFirstStrategy (
    const CustomVector< Student > & )
```

5.36.1.7 sortNuskriaustiAndProtingiFirstStrategy() [2/5]

```
template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiFirstStrategy (
    const std::deque< Student > & )
```

5.36.1.8 sortNuskriaustiAndProtingiFirstStrategy() [3/5]

```
template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiFirstStrategy (
    const std::list< Student > & )
```

5.36.1.9 sortNuskriaustiAndProtingiFirstStrategy() [4/5]

```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiFirstStrategy (
    const T & students )
```

5.36.1.10 sortNuskriaustiAndProtingiFirstStrategy() [5/5]

```
template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiFirstStrategy (
    const vector< Student > & )
```

5.36.1.11 sortNuskriaustiAndProtingiSecondStrategy() [1/5]

```
template CustomVector< Student > sortNuskriaustiAndProtingiSecondStrategy (
    CustomVector< Student > & )
```

5.36.1.12 sortNuskriaustiAndProtingiSecondStrategy() [2/5]

```
template std::deque< Student > sortNuskriaustiAndProtingiSecondStrategy (
    std::deque< Student > & )
```

5.36.1.13 sortNuskriaustiAndProtingiSecondStrategy() [3/5]

```
template std::list< Student > sortNuskriaustiAndProtingiSecondStrategy (
    std::list< Student > & )
```

5.36.1.14 sortNuskriaustiAndProtingiSecondStrategy() [4/5]

```
template<typename T >
T sortNuskriaustiAndProtingiSecondStrategy (
    T & students )
```

5.36.1.15 sortNuskriaustiAndProtingiSecondStrategy() [5/5]

```
template vector< Student > sortNuskriaustiAndProtingiSecondStrategy (
    vector< Student > & )
```

5.36.1.16 sortNuskriaustiAndProtingiThirdStrategy() [1/5]

```
template std::pair< CustomVector< Student >, CustomVector< Student > > sortNuskriaustiAndProtingiThirdStrategy (
    CustomVector< Student > & )
```

5.36.1.17 sortNuskriaustiAndProtingiThirdStrategy() [2/5]

```
template std::pair< std::deque< Student >, std::deque< Student > > sortNuskriaustiAndProtingiThirdStrategy (
    std::deque< Student > & )
```

5.36.1.18 sortNuskriaustiAndProtingiThirdStrategy() [3/5]

```
template std::pair< std::list< Student >, std::list< Student > > sortNuskriaustiAndProtingiThirdStrategy (
    std::list< Student > & )
```

5.36.1.19 sortNuskriaustiAndProtingiThirdStrategy() [4/5]

```
template<typename T >
std::pair< T, T > sortNuskriaustiAndProtingiThirdStrategy (
    T & students )
```

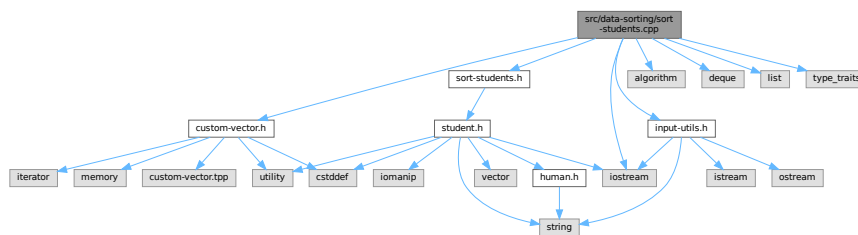
5.36.1.20 sortNuskriaustiAndProtingiThirdStrategy() [5/5]

```
template std::pair< vector< Student >, vector< Student > > sortNuskriaustiAndProtingiThirdStrategy (
    vector< Student > & )
```

5.37 src/data-sorting/sort-students.cpp File Reference

```
#include "sort-students.h"
#include "custom-vector.h"
#include "input-utils.h"
#include <algorithm>
#include <iostream>
#include <deque>
#include <list>
#include <type_traits>
```

Include dependency graph for sort-students.cpp:



Functions

- template<typename T>
void sortStudents (T &students, int sortChoice, bool ascending)
- template void sortStudents (vector< Student > &, int, bool)
- template void sortStudents (std::list< Student > &, int, bool)
- template void sortStudents (std::deque< Student > &, int, bool)
- template void sortStudents (CustomVector< Student > &, int, bool)
- template<typename T>
void runSortStudentsChoicePrompt (T &students)
- template void runSortStudentsChoicePrompt (vector< Student > &)
- template void runSortStudentsChoicePrompt (std::list< Student > &)
- template void runSortStudentsChoicePrompt (std::deque< Student > &)
- template void runSortStudentsChoicePrompt (CustomVector< Student > &)

5.37.1 Function Documentation

5.37.1.1 runSortStudentsChoicePrompt() [1/5]

```
template void runSortStudentsChoicePrompt (
    CustomVector< Student > & )
```

5.37.1.2 runSortStudentsChoicePrompt() [2/5]

```
template void runSortStudentsChoicePrompt (
    std::deque< Student > & )
```

5.37.1.3 runSortStudentsChoicePrompt() [3/5]

```
template void runSortStudentsChoicePrompt (
    std::list< Student > & )
```

5.37.1.4 runSortStudentsChoicePrompt() [4/5]

```
template<typename T >
void runSortStudentsChoicePrompt (
    T & students )
```

5.37.1.5 runSortStudentsChoicePrompt() [5/5]

```
template void runSortStudentsChoicePrompt (
    vector< Student > & )
```

5.37.1.6 sortStudents() [1/5]

```
template void sortStudents (
    CustomVector< Student > & ,
    int ,
    bool )
```

5.37.1.7 sortStudents() [2/5]

```
template void sortStudents (
    std::deque< Student > & ,
    int ,
    bool )
```

5.37.1.8 sortStudents() [3/5]

```
template void sortStudents (
    std::list< Student > & ,
    int ,
    bool )
```


5.37.1.9 sortStudents() [4/5]

```
template<typename T >
void sortStudents (
    T & students,
    int sortChoice,
    bool ascending )
```

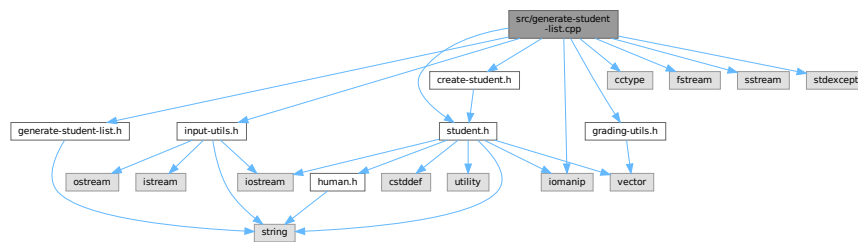
5.37.1.10 sortStudents() [5/5]

```
template void sortStudents (
    vector< Student > & ,
    int ,
    bool )
```

5.38 src/generate-student-list.cpp File Reference

```
#include "generate-student-list.h"
#include "input-utils.h"
#include "student.h"
#include "create-student.h"
#include "grading-utils.h"
#include <cctype>
#include <fstream>
#include <iomanip>
#include <sstream>
#include <stdexcept>
```

Include dependency graph for generate-student-list.cpp:



Functions

- void [printHeader](#) (std::ostream &output)
- void [printStudentListRow](#) (std::ostream &output, const [Student](#) &s)
- void [generateStudentListFile](#) (int studentCount, const std::string &filename)
- void [handleGenerateStudentListFile](#) ()

5.38.1 Function Documentation

5.38.1.1 generateStudentListFile()

```
void generateStudentListFile (
    int studentCount,
    const std::string & filename )
```

5.38.1.2 handleGenerateStudentListFile()

```
void handleGenerateStudentListFile ( )
```

5.38.1.3 printHeader()

```
void printHeader (
    std::ostream & output )
```

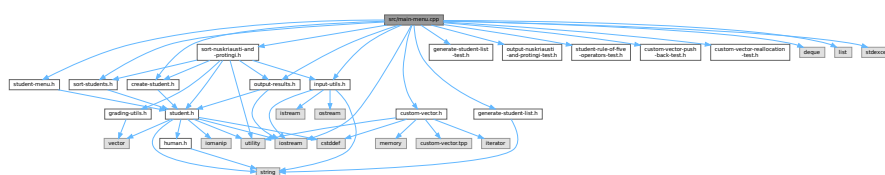
5.38.1.4 printStudentListRow()

```
void printStudentListRow (
    std::ostream & output,
    const Student & s )
```

5.39 src/main-menu.cpp File Reference

```
#include "student-menu.h"
#include "custom-vector.h"
#include "create-student.h"
#include "input-utils.h"
#include "output-results.h"
#include "generate-student-list.h"
#include "generate-student-list-test.h"
#include "output-nuskriausti-and-protingi-test.h"
#include "student-rule-of-five-operators-test.h"
#include "custom-vector-push-back-test.h"
#include "custom-vector-reallocation-test.h"
#include "sort-nuskriausti-and-protingi.h"
#include "sort-students.h"
#include <iostream>
#include <deque>
#include <list>
#include <stdexcept>
```

Include dependency graph for main-menu.cpp:



Functions

- void `showMainMenu` ()
- template<typename T >
void `handleMenuChoice` (int choice, T &students)
- template void `handleMenuChoice`< vector< Student > > (int, vector< Student > &)
- template void `handleMenuChoice`< std::list< Student > > (int, std::list< Student > &)
- template void `handleMenuChoice`< std::deque< Student > > (int, std::deque< Student > &)
- template void `handleMenuChoice`< CustomVector< Student > > (int, CustomVector< Student > &)
- template<typename T >
void `startMainMenu` ()
- template void `startMainMenu`< vector< Student > > ()
- template void `startMainMenu`< std::list< Student > > ()
- template void `startMainMenu`< std::deque< Student > > ()
- template void `startMainMenu`< CustomVector< Student > > ()

5.39.1 Function Documentation

5.39.1.1 `handleMenuChoice()`

```
template<typename T >
void handleMenuChoice (
    int choice,
    T & students )
```

5.39.1.2 `handleMenuChoice< CustomVector< Student > >()`

```
template void handleMenuChoice< CustomVector< Student > > (
    int ,
    CustomVector< Student > & )
```

5.39.1.3 `handleMenuChoice< std::deque< Student > >()`

```
template void handleMenuChoice< std::deque< Student > > (
    int ,
    std::deque< Student > & )
```

5.39.1.4 `handleMenuChoice< std::list< Student > >()`

```
template void handleMenuChoice< std::list< Student > > (
    int ,
    std::list< Student > & )
```

5.39.1.5 `handleMenuChoice< vector< Student > >()`

```
template void handleMenuChoice< vector< Student > > (
    int ,
    vector< Student > & )
```

5.39.1.6 showMainMenu()

```
void showMainMenu ( )
```

5.39.1.7 startMainMenu()

```
template<typename T >
void startMainMenu ( )
```

5.39.1.8 startMainMenu< CustomVector< Student > >()

```
template void startMainMenu< CustomVector< Student > > ( )
```

5.39.1.9 startMainMenu< std::deque< Student > >()

```
template void startMainMenu< std::deque< Student > > ( )
```

5.39.1.10 startMainMenu< std::list< Student > >()

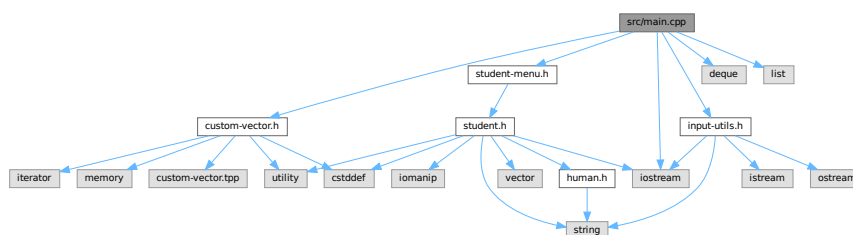
```
template void startMainMenu< std::list< Student > > ( )
```

5.39.1.11 startMainMenu< vector< Student > >()

```
template void startMainMenu< vector< Student > > ( )
```

5.40 src/main.cpp File Reference

```
#include "student-menu.h"
#include "input-utils.h"
#include "custom-vector.h"
#include <deque>
#include <iostream>
#include <list>
Include dependency graph for main.cpp:
```



Functions

- int [main](#) ()

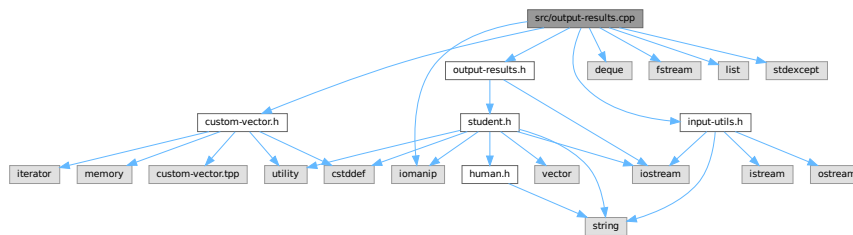
5.40.1 Function Documentation

5.40.1.1 main()

```
int main ( )
```

5.41 src/output-results.cpp File Reference

```
#include "output-results.h"
#include "custom-vector.h"
#include "input-utils.h"
#include <deque>
#include <fstream>
#include <iomanip>
#include <list>
#include <stdexcept>
Include dependency graph for output-results.cpp:
```



Functions

- template<typename T >
void [outputResults](#) (const T &students, std::ostream &out)
- template void [outputResults](#) (const vector< [Student](#) > &students, std::ostream &out)
- template void [outputResults](#) (const std::list< [Student](#) > &students, std::ostream &out)
- template void [outputResults](#) (const std::deque< [Student](#) > &students, std::ostream &out)
- template void [outputResults](#) (const [CustomVector](#)< [Student](#) > &students, std::ostream &out)
- template<typename T >
void [chooseOutputAndPrint](#) (T &students)
- template void [chooseOutputAndPrint](#) (vector< [Student](#) > &students)
- template void [chooseOutputAndPrint](#) (std::list< [Student](#) > &students)
- template void [chooseOutputAndPrint](#) (std::deque< [Student](#) > &students)
- template void [chooseOutputAndPrint](#) ([CustomVector](#)< [Student](#) > &students)

5.41.1 Function Documentation

5.41.1.1 chooseOutputAndPrint() [1/5]

```
template void chooseOutputAndPrint (
    CustomVector< Student > & students )
```

5.41.1.2 chooseOutputAndPrint() [2/5]

```
template void chooseOutputAndPrint (
    std::deque< Student > & students )
```

5.41.1.3 chooseOutputAndPrint() [3/5]

```
template void chooseOutputAndPrint (
    std::list< Student > & students )
```

5.41.1.4 chooseOutputAndPrint() [4/5]

```
template<typename T >
void chooseOutputAndPrint (
    T & students )
```

5.41.1.5 chooseOutputAndPrint() [5/5]

```
template void chooseOutputAndPrint (
    vector< Student > & students )
```

5.41.1.6 outputResults() [1/5]

```
template void outputResults (
    const CustomVector< Student > & students,
    std::ostream & out )
```

5.41.1.7 outputResults() [2/5]

```
template void outputResults (
    const std::deque< Student > & students,
    std::ostream & out )
```

5.41.1.8 outputResults() [3/5]

```
template void outputResults (
    const std::list< Student > & students,
    std::ostream & out )
```

5.41.1.9 `outputResults()` [4/5]

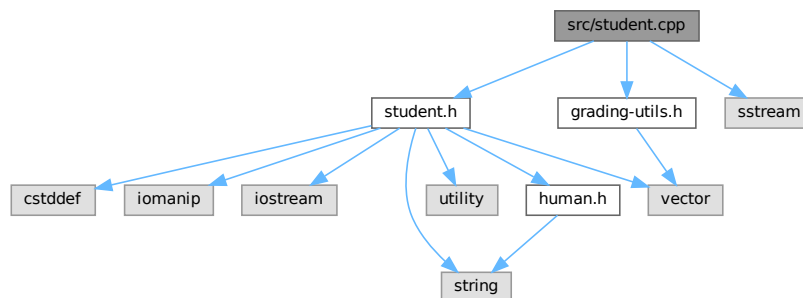
```
template<typename T >
void outputResults (
    const T & students,
    std::ostream & out )
```

5.41.1.10 `outputResults()` [5/5]

```
template void outputResults (
    const vector< Student > & students,
    std::ostream & out )
```

5.42 `src/student.cpp` File Reference

```
#include "student.h"
#include "grading-utils.h"
#include <sstream>
Include dependency graph for student.cpp:
```



Functions

- `std::istream & operator>>` (`std::istream &is`, `Student &student`)
- `std::ostream & operator<<` (`std::ostream &os`, `const Student &student`)

5.42.1 Function Documentation

5.42.1.1 `operator<<()`

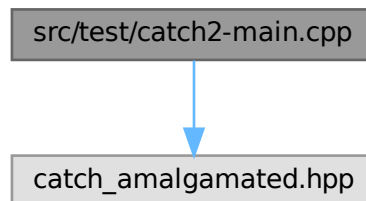
```
std::ostream & operator<< (
    std::ostream & os,
    const Student & student )
```

5.42.1.2 operator>>()

```
std::istream & operator>> (
    std::istream & is,
    Student & student )
```

5.43 src/test/catch2-main.cpp File Reference

```
#include "catch_amalgamated.hpp"
Include dependency graph for catch2-main.cpp:
```



Macros

- #define CATCH_CONFIG_MAIN

5.43.1 Macro Definition Documentation

5.43.1.1 CATCH_CONFIG_MAIN

```
#define CATCH_CONFIG_MAIN
```

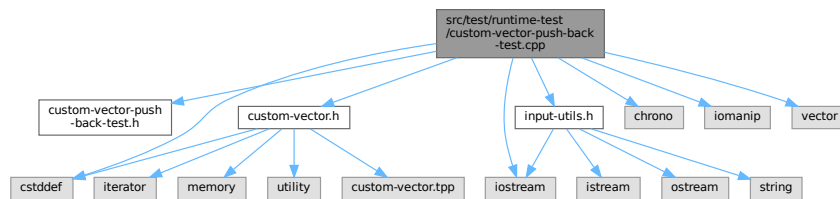
5.44 src/test/runtime-test/custom-vector-push-back-test.cpp File Reference

```
#include "custom-vector-push-back-test.h"
#include "custom-vector.h"
#include "input-utils.h"
#include <chrono>
#include <cstdint>
#include <iomanip>
#include <iostream>
```



```
#include <vector>
```

Include dependency graph for custom-vector-push-back-test.cpp:



Functions

- void [handleCustomVectorPushBackTest](#) ()

5.44.1 Function Documentation

5.44.1.1 handleCustomVectorPushBackTest()

```
void handleCustomVectorPushBackTest ( )
```

5.45 src/test/runtime-test/custom-vector-reallocation-test.cpp File Reference

```
#include "custom-vector-reallocation-test.h"
```

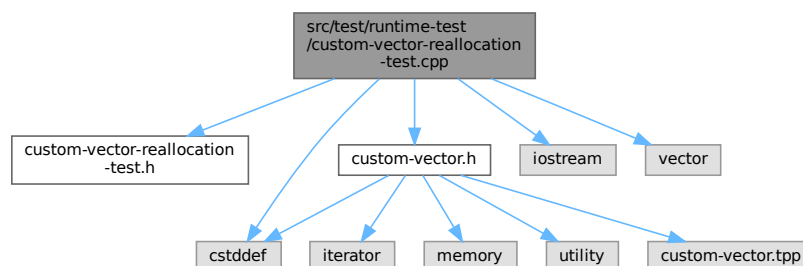
```
#include "custom-vector.h"
```

```
#include <cstddef>
```

```
#include <iostream>
```

```
#include <vector>
```

Include dependency graph for custom-vector-reallocation-test.cpp:



Functions

- template<typename Vector >
std::size_t [countReallocationsWhilePushBack](#) (std::size_t element_count)
- void [handleCustomVectorReallocationTest](#) ()

Variables

- constexpr std::size_t `kElementCount` = 100'000'000

5.45.1 Function Documentation

5.45.1.1 countReallocationsWhilePushBack()

```
template<typename Vector >
std::size_t countReallocationsWhilePushBack (
    std::size_t element_count )
```

5.45.1.2 handleCustomVectorReallocationTest()

```
void handleCustomVectorReallocationTest ( )
```

5.45.2 Variable Documentation

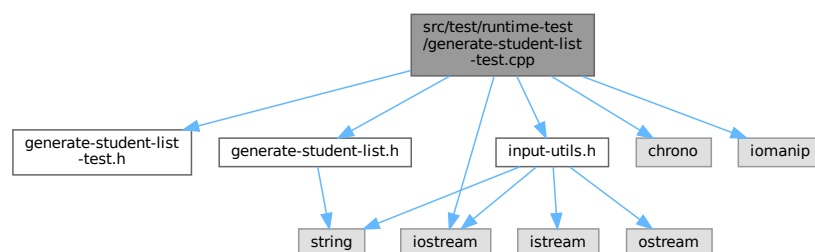
5.45.2.1 kElementCount

```
constexpr std::size_t kElementCount = 100'000'000 [constexpr]
```

5.46 src/test/runtime-test/generate-student-list-test.cpp File Reference

```
#include "generate-student-list-test.h"
#include "generate-student-list.h"
#include "input-utils.h"
#include <chrono>
#include <iomanip>
#include <iostream>
```

Include dependency graph for generate-student-list-test.cpp:



Functions

- void `handleGenerateStudentListTest` ()

5.46.1 Function Documentation

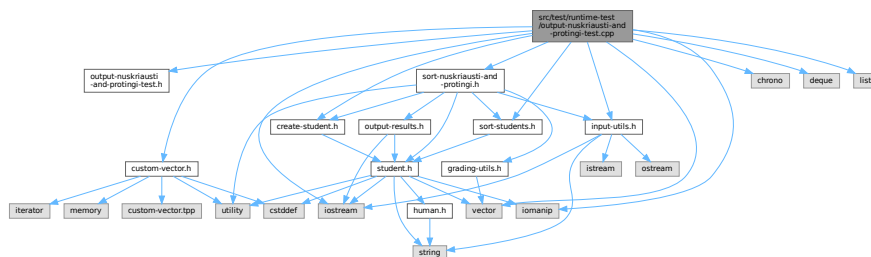
5.46.1.1 handleGenerateStudentListTest()

```
void handleGenerateStudentListTest ( )
```

5.47 src/test/runtime-test/output-nuskriausti-and-protingi-test.cpp File Reference

```
#include "output-nuskriausti-and-protingi-test.h"
#include "custom-vector.h"
#include "input-utils.h"
#include "create-student.h"
#include "sort-students.h"
#include "sort-nuskriausti-and-protingi.h"
#include <chrono>
#include <deque>
#include <iomanip>
#include <iostream>
#include <list>
#include <vector>
```

Include dependency graph for output-nuskriausti-and-protingi-test.cpp:



Functions

- `template<typename Container >`
void `outputNuskriaustiAndProtingiTestForContainer` (const string &filename, int strategyChoice)
- void `outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles` ()

5.47.1 Function Documentation

5.47.1.1 outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles()

```
void outputNuskriaustiAndProtingiTestForAllStudentsTxtFiles ( )
```

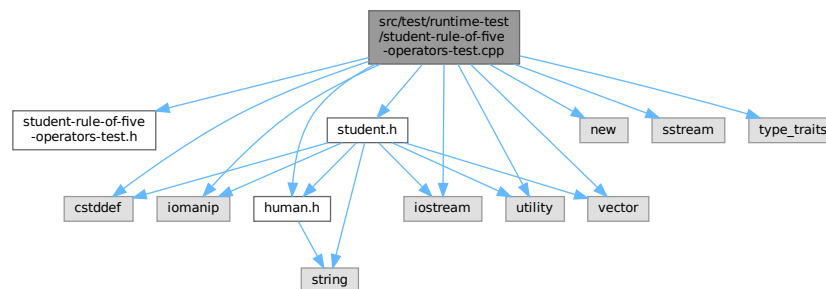
5.47.1.2 outputNuskriaustiAndProtingiTestForContainer()

```
template<typename Container >
void outputNuskriaustiAndProtingiTestForContainer (
    const string & filename,
    int strategyChoice )
```

5.48 src/test/runtime-test/student-rule-of-five-operators-test.cpp File Reference

```
#include "student-rule-of-five-operators-test.h"
#include "human.h"
#include "student.h"
#include <cstddef>
#include <iomanip>
#include <iostream>
#include <new>
#include <sstream>
#include <type_traits>
#include <utility>
#include <vector>
```

Include dependency graph for student-rule-of-five-operators-test.cpp:



Functions

- void [printTestResult](#) (const char *testName, bool passed)
- bool [testCopyConstructor](#) ()
- bool [testMoveConstructor](#) ()
- bool [testCopyAssignment](#) ()
- bool [testMoveAssignment](#) ()
- bool [testDestructor](#) ()
- bool [testInputOperatorValid](#) ()
- bool [testInputOperatorInvalid](#) ()
- bool [testOutputOperator](#) ()
- bool [testHumanIsAbstractAndNotInstantiable](#) ()
- void [handleStudentRuleOfFiveAndOperatorsTest](#) ()

5.48.1 Function Documentation

5.48.1.1 `handleStudentRuleOfFiveAndOperatorsTest()`

```
void handleStudentRuleOfFiveAndOperatorsTest ( )
```

5.48.1.2 `printTestResult()`

```
void printTestResult (
    const char * testName,
    bool passed )
```

5.48.1.3 `testCopyAssignment()`

```
bool testCopyAssignment ( )
```

5.48.1.4 `testCopyConstructor()`

```
bool testCopyConstructor ( )
```

5.48.1.5 `testDestructor()`

```
bool testDestructor ( )
```

5.48.1.6 `testHumanIsAbstractAndNotInstantiable()`

```
bool testHumanIsAbstractAndNotInstantiable ( )
```

5.48.1.7 `testInputOperatorInvalid()`

```
bool testInputOperatorInvalid ( )
```

5.48.1.8 `testInputOperatorValid()`

```
bool testInputOperatorValid ( )
```

5.48.1.9 `testMoveAssignment()`

```
bool testMoveAssignment ( )
```

5.48.1.10 testMoveConstructor()

```
bool testMoveConstructor ( )
```

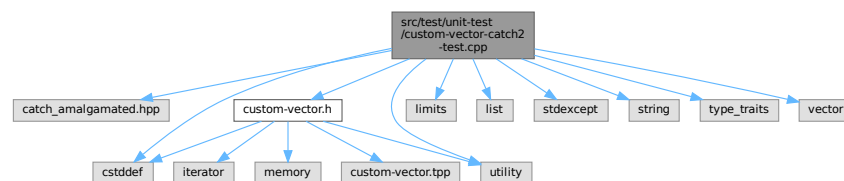
5.48.1.11 testOutputOperator()

```
bool testOutputOperator ( )
```

5.49 src/test/unit-test/custom-vector-catch2-test.cpp File Reference

```
#include "catch_amalgamated.hpp"
#include "custom-vector.h"
#include <cstdlib>
#include <limits>
#include <list>
#include <stdexcept>
#include <string>
#include <type_traits>
#include <utility>
#include <vector>
```

Include dependency graph for custom-vector-catch2-test.cpp:



Functions

- **TEST_CASE** ("CustomVector default constructor creates empty vector", "[custom-vector][constructors]")
- **TEST_CASE** ("CustomVector copy constructor performs deep copy", "[custom-vector][constructors]")
- **TEST_CASE** ("CustomVector move constructor transfers ownership", "[custom-vector][constructors]")
- **TEST_CASE** ("CustomVector copy assignment performs deep copy", "[custom-vector][assignment]")
- **TEST_CASE** ("CustomVector copy assignment handles self-assignment", "[custom-vector][assignment]")
- **TEST_CASE** ("CustomVector move assignment transfers ownership", "[custom-vector][assignment]")
- **TEST_CASE** ("CustomVector swap exchanges contents", "[custom-vector][swap]")
- **TEST_CASE** ("CustomVector assign replaces contents", "[custom-vector][assign]")
- **TEST_CASE** ("CustomVector insert inserts at valid positions", "[custom-vector][insert]")
- **TEST_CASE** ("CustomVector insert throws on invalid position", "[custom-vector][insert]")
- **TEST_CASE** ("CustomVector insert_range inserts multiple elements", "[custom-vector][insert]")
- **TEST_CASE** ("CustomVector insert_range throws on invalid position", "[custom-vector][insert]")
- **TEST_CASE** ("CustomVector emplace constructs element in place", "[custom-vector][emplace]")
- **TEST_CASE** ("CustomVector emplace throws on invalid position", "[custom-vector][emplace]")
- **TEST_CASE** ("CustomVector push_back and emplace_back append elements", "[custom-vector][modifiers]")
- **TEST_CASE** ("CustomVector pop_back removes last element", "[custom-vector][modifiers]")
- **TEST_CASE** ("CustomVector append_range appends elements at end", "[custom-vector][modifiers]")

- `TEST_CASE` ("CustomVector erase removes single element", "[custom-vector][erase]")
- `TEST_CASE` ("CustomVector erase removes element range", "[custom-vector][erase]")
- `TEST_CASE` ("CustomVector erase returns first on empty range", "[custom-vector][erase]")
- `TEST_CASE` ("CustomVector erase throws on invalid range", "[custom-vector][erase]")
- `TEST_CASE` ("CustomVector element access provides references", "[custom-vector][access]")
- `TEST_CASE` ("CustomVector at throws on out of range index", "[custom-vector][access]")
- `TEST_CASE` ("CustomVector capacity and size track elements", "[custom-vector][capacity]")
- `TEST_CASE` ("CustomVector reserve grows capacity and preserves elements", "[custom-vector][capacity]")
- `TEST_CASE` ("CustomVector resize grows and shrinks", "[custom-vector][capacity]")
- `TEST_CASE` ("CustomVector resize throws when count exceeds max_size", "[custom-vector][capacity]")
- `TEST_CASE` ("CustomVector shrink_to_fit reduces capacity to size", "[custom-vector][capacity]")
- `TEST_CASE` ("CustomVector clear removes all elements", "[custom-vector][capacity]")
- `TEST_CASE` ("CustomVector iterators and data expose element storage", "[custom-vector][iterators]")
- `TEST_CASE` ("CustomVector comparison operators compare lexicographically", "[custom-vector][compare]")

5.49.1 Function Documentation

5.49.1.1 `TEST_CASE()` [1/31]

```
TEST_CASE (
    "CustomVector append_range appends elements at end" ,
    "" [custom-vector][modifiers] )
```

5.49.1.2 `TEST_CASE()` [2/31]

```
TEST_CASE (
    "CustomVector assign replaces contents" ,
    "" [custom-vector][assign] )
```

5.49.1.3 `TEST_CASE()` [3/31]

```
TEST_CASE (
    "CustomVector at throws on out of range index" ,
    "" [custom-vector][access] )
```

5.49.1.4 `TEST_CASE()` [4/31]

```
TEST_CASE (
    "CustomVector capacity and size track elements" ,
    "" [custom-vector][capacity] )
```

5.49.1.5 `TEST_CASE()` [5/31]

```
TEST_CASE (
    "CustomVector clear removes all elements" ,
    "" [custom-vector][capacity] )
```

5.49.1.6 TEST_CASE() [6/31]

```
TEST_CASE (
    "CustomVector comparison operators compare lexicographically" ,
    "" [custom-vector][compare] )
```

5.49.1.7 TEST_CASE() [7/31]

```
TEST_CASE (
    "CustomVector copy assignment handles self-assignment" ,
    "" [custom-vector][assignment] )
```

5.49.1.8 TEST_CASE() [8/31]

```
TEST_CASE (
    "CustomVector copy assignment performs deep copy" ,
    "" [custom-vector][assignment] )
```

5.49.1.9 TEST_CASE() [9/31]

```
TEST_CASE (
    "CustomVector copy constructor performs deep copy" ,
    "" [custom-vector][constructors] )
```

5.49.1.10 TEST_CASE() [10/31]

```
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    "" [custom-vector][constructors] )
```

5.49.1.11 TEST_CASE() [11/31]

```
TEST_CASE (
    "CustomVector element access provides references" ,
    "" [custom-vector][access] )
```

5.49.1.12 TEST_CASE() [12/31]

```
TEST_CASE (
    "CustomVector emplace constructs element in place" ,
    "" [custom-vector][emplace] )
```

5.49.1.13 TEST_CASE() [13/31]

```
TEST_CASE (
    "CustomVector emplace throws on invalid position" ,
    "" [custom-vector][emplace] )
```


5.49.1.14 TEST_CASE() [14/31]

```
TEST_CASE (
    "CustomVector erase removes element range" ,
    "" [custom-vector][erase] )
```

5.49.1.15 TEST_CASE() [15/31]

```
TEST_CASE (
    "CustomVector erase removes single element" ,
    "" [custom-vector][erase] )
```

5.49.1.16 TEST_CASE() [16/31]

```
TEST_CASE (
    "CustomVector erase returns first on empty range" ,
    "" [custom-vector][erase] )
```

5.49.1.17 TEST_CASE() [17/31]

```
TEST_CASE (
    "CustomVector erase throws on invalid range" ,
    "" [custom-vector][erase] )
```

5.49.1.18 TEST_CASE() [18/31]

```
TEST_CASE (
    "CustomVector insert inserts at valid positions" ,
    "" [custom-vector][insert] )
```

5.49.1.19 TEST_CASE() [19/31]

```
TEST_CASE (
    "CustomVector insert throws on invalid position" ,
    "" [custom-vector][insert] )
```

5.49.1.20 TEST_CASE() [20/31]

```
TEST_CASE (
    "CustomVector insert_range inserts multiple elements" ,
    "" [custom-vector][insert] )
```

5.49.1.21 TEST_CASE() [21/31]

```
TEST_CASE (
    "CustomVector insert_range throws on invalid position" ,
    "" [custom-vector][insert] )
```

5.49.1.22 TEST_CASE() [22/31]

```
TEST_CASE (
    "CustomVector iterators and data expose element storage" ,
    "" [custom-vector][iterators] )
```

5.49.1.23 TEST_CASE() [23/31]

```
TEST_CASE (
    "CustomVector move assignment transfers ownership" ,
    "" [custom-vector][assignment] )
```

5.49.1.24 TEST_CASE() [24/31]

```
TEST_CASE (
    "CustomVector move constructor transfers ownership" ,
    "" [custom-vector][constructors] )
```

5.49.1.25 TEST_CASE() [25/31]

```
TEST_CASE (
    "CustomVector pop_back removes last element" ,
    "" [custom-vector][modifiers] )
```

5.49.1.26 TEST_CASE() [26/31]

```
TEST_CASE (
    "CustomVector push_back and emplace_back append elements" ,
    "" [custom-vector][modifiers] )
```

5.49.1.27 TEST_CASE() [27/31]

```
TEST_CASE (
    "CustomVector reserve grows capacity and preserves elements" ,
    "" [custom-vector][capacity] )
```

5.49.1.28 TEST_CASE() [28/31]

```
TEST_CASE (
    "CustomVector resize grows and shrinks" ,
    "" [custom-vector][capacity] )
```

5.49.1.29 TEST_CASE() [29/31]

```
TEST_CASE (
    "CustomVector resize throws when count exceeds max_size" ,
    "" [custom-vector][capacity] )
```

5.49.1.30 TEST_CASE() [30/31]

```
TEST_CASE (
    "CustomVector shrink_to_fit reduces capacity to size" ,
    "" [custom-vector][capacity] )
```

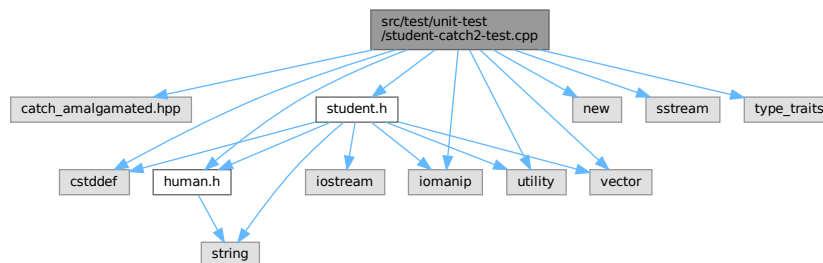
5.49.1.31 TEST_CASE() [31/31]

```
TEST_CASE (
    "CustomVector swap exchanges contents" ,
    "" [custom-vector][swap] )
```

5.50 src/test/unit-test/student-catch2-test.cpp File Reference

```
#include "catch_amalgamated.hpp"
#include "human.h"
#include "student.h"
#include <cstddef>
#include <iomanip>
#include <new>
#include <sstream>
#include <type_traits>
#include <utility>
#include <vector>
```

Include dependency graph for student-catch2-test.cpp:



Functions

- [TEST_CASE](#) ("Student copy constructor copies all fields", "[student][rule-of-five]")
- [TEST_CASE](#) ("Student move constructor moves all fields and resets source", "[student][rule-of-five]")
- [TEST_CASE](#) ("Student copy assignment copies all fields", "[student][rule-of-five]")
- [TEST_CASE](#) ("Student move assignment moves all fields and resets source", "[student][rule-of-five]")
- [TEST_CASE](#) ("Student destructor allows safe destruction and reconstruction in place", "[student][rule-of-five]")
- [TEST_CASE](#) ("Student satisfies basic rule-of-five traits", "[student][rule-of-five]")
- [TEST_CASE](#) ("Student input operator parses valid input", "[student][operators]")
- [TEST_CASE](#) ("Student input operator fails on invalid input", "[student][operators]")
- [TEST_CASE](#) ("Student output operator formats output correctly", "[student][operators]")
- [TEST_CASE](#) ("Human is abstract and not default constructible", "[human][type-traits]")

5.50.1 Function Documentation

5.50.1.1 TEST_CASE() [1/10]

```
TEST_CASE (
    "Human is abstract and not default constructible" ,
    "" [human][type-traits] )
```

5.50.1.2 TEST_CASE() [2/10]

```
TEST_CASE (
    "Student copy assignment copies all fields" ,
    "" [student][rule-of-five] )
```

5.50.1.3 TEST_CASE() [3/10]

```
TEST_CASE (
    "Student copy constructor copies all fields" ,
    "" [student][rule-of-five] )
```

5.50.1.4 TEST_CASE() [4/10]

```
TEST_CASE (
    "Student destructor allows safe destruction and reconstruction in place" ,
    "" [student][rule-of-five] )
```

5.50.1.5 TEST_CASE() [5/10]

```
TEST_CASE (
    "Student input operator fails on invalid input" ,
    "" [student][operators] )
```

5.50.1.6 TEST_CASE() [6/10]

```
TEST_CASE (
    "Student input operator parses valid input" ,
    "" [student][operators] )
```

5.50.1.7 TEST_CASE() [7/10]

```
TEST_CASE (
    "Student move assignment moves all fields and resets source" ,
    "" [student][rule-of-five] )
```

5.50.1.8 TEST_CASE() [8/10]

```
TEST_CASE (
    "Student move constructor moves all fields and resets source" ,
    "" [student][rule-of-five] )
```

5.50.1.9 TEST_CASE() [9/10]

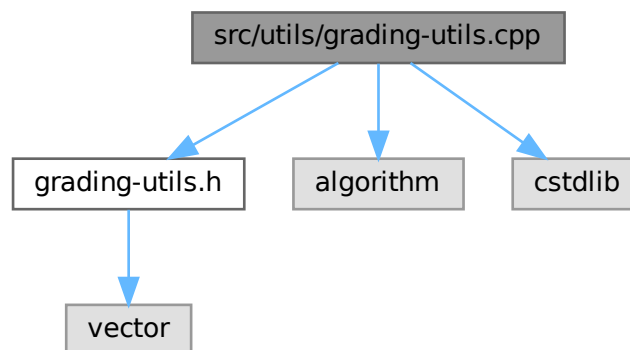
```
TEST_CASE (
    "Student output operator formats output correctly" ,
    "" [student][operators] )
```

5.50.1.10 TEST_CASE() [10/10]

```
TEST_CASE (
    "Student satisfies basic rule-of-five traits" ,
    "" [student][rule-of-five] )
```

5.51 src/utls/grading-utils.cpp File Reference

```
#include "grading-utils.h"
#include <algorithm>
#include <cstdlib>
Include dependency graph for grading-utils.cpp:
```



Functions

- int [randomGrade](#) (int min, int max)
- double [calculateFinalGradeMedian](#) (const vector< int > &homeworkGrades, int examGrade)
- double [calculateFinalGradeAverage](#) (const vector< int > &homeworkGrades, int examGrade)

5.51.1 Function Documentation

5.51.1.1 calculateFinalGradeAverage()

```
double calculateFinalGradeAverage (
    const vector< int > & homeworkGrades,
    int examGrade )
```

5.51.1.2 calculateFinalGradeMedian()

```
double calculateFinalGradeMedian (
    const vector< int > & homeworkGrades,
    int examGrade )
```

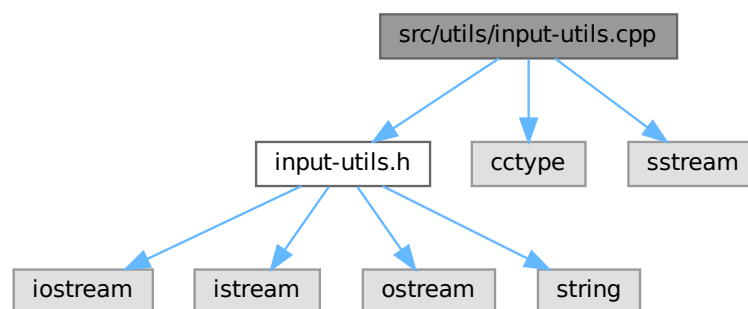
5.51.1.3 randomGrade()

```
int randomGrade (
    int min,
    int max )
```

5.52 src/utls/input-utls.cpp File Reference

```
#include "input-utls.h"
#include <cctype>
#include <sstream>
```

Include dependency graph for input-utls.cpp:



Functions

- int [readIntInRange](#) (const string &prompt, int min, int max)
- string [readSingleStringToken](#) (const string &prompt)
- char [readYesOrNo](#) (const string &prompt)
- int [readSingleIntToken](#) (const string &prompt)

5.52.1 Function Documentation

5.52.1.1 readIntInRange()

```
int readIntInRange (
    const string & prompt,
    int min,
    int max )
```

5.52.1.2 readSingleIntToken()

```
int readSingleIntToken (
    const string & prompt )
```

5.52.1.3 readSingleStringToken()

```
string readSingleStringToken (
    const string & prompt )
```

5.52.1.4 readYesOrNo()

```
char readYesOrNo (
    const string & prompt )
```


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